21CSC529T Inferential Statistics

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Agenda

- Covariance
- Correlation
- Permutation and Combination
- Types of Probability
- Rules for computing probability
- Marginal probability
- Conditional probability
- Bayes Theorem
- Applications of Bayes Theorem
- Problem-solving in Bayes theorem

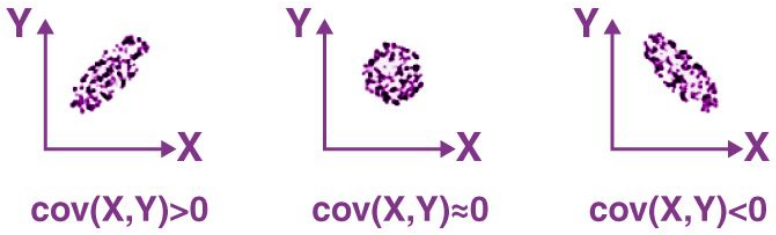
This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Covariance

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Covariance

- Covariance is a measure of how two variables change together. It indicates the direction of the linear relationship between the variables, but it doesn't provide the strength or magnitude of the relationship in a standardized way.
- Covariance is similar to variance, but variance explains how a single variable varies, and covariance explains how two variables vary together.
- It cannot tell whether the value indicates a strong relationship or weak relationship, since the covariance can take any value
- Covariance can theoretically range from $-\infty$ to $+\infty$.



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Covariance

The covariance for two variables X and Y are given by

$$\text{Population Covariance: } \mathrm{Cov}(x,\ y) = \frac{\sum_{i=1}^{N}\left(X_i - \mu_x\right)\left(Y_i - \mu_Y\right)}{N}$$

$$\text{Sample Covariance: } \mathrm{Cov}(x,\ y) = \frac{\sum_{i=1}^{N}\left(X_i - \bar{x}\right)\left(Y_i - \bar{y}\right)}{n-1}$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Suppose we have marks of 5 students in **Math** and **Science**. Find the relation between the mark scoring ability of the students.

Student	Math (X)	Science (Y)
1	80	85
2	90	95
3	70	65
4	60	70
5	85	80

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Student	X	Y	$X-\bar{X}$	$Y-\bar{Y}$	$(X-\bar{X})(Y-\bar{Y})$
1	80	85	3	6	18
2	90	95	13	16	208
3	70	65	-7	-14	98
4	60	70	-17	-9	153
5	85	80	8	1	8
Σ					**485**

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\mathrm{Cov}(X,Y) = \frac{\sum (X-\bar{X})(Y-\bar{Y})}{n}$$

$$\mathrm{Cov}(X,Y) = \frac{485}{5} = 97$$

Positive covariance (97) → Math and Science marks tend to increase together.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Correlation

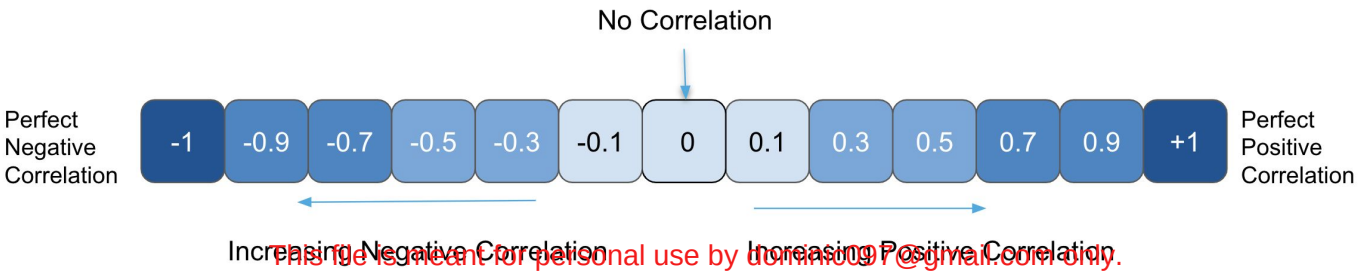
This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Correlation

- Correlation is a statistical measure that describes the strength and direction of the linear relationship between two variables.
- Unlike covariance, which is in units derived from the original variables, correlation is a standardized measure and ranges from −1 to +1.

- It indicates the extent to which two variables increase or decrease in parallel

- The value of a correlation coefficient ranges between -1 and 1



This file is meant for personal use by dominic007@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Correlation

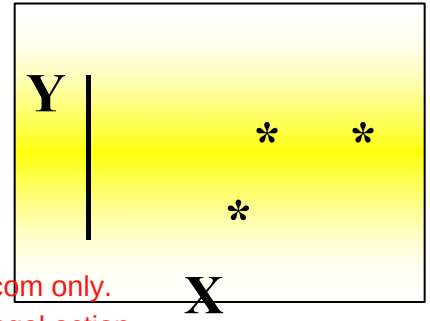
Finding the relationship between two quantitative variables without being able to infer causal relationships

Correlation is a statistical technique used to determine the degree to which two variables are related

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Scatter diagram

- Rectangular coordinate
- Two quantitative variables
- One variable is called independent (X) and the second is called dependent (Y)
- Points are not joined
- No frequency table

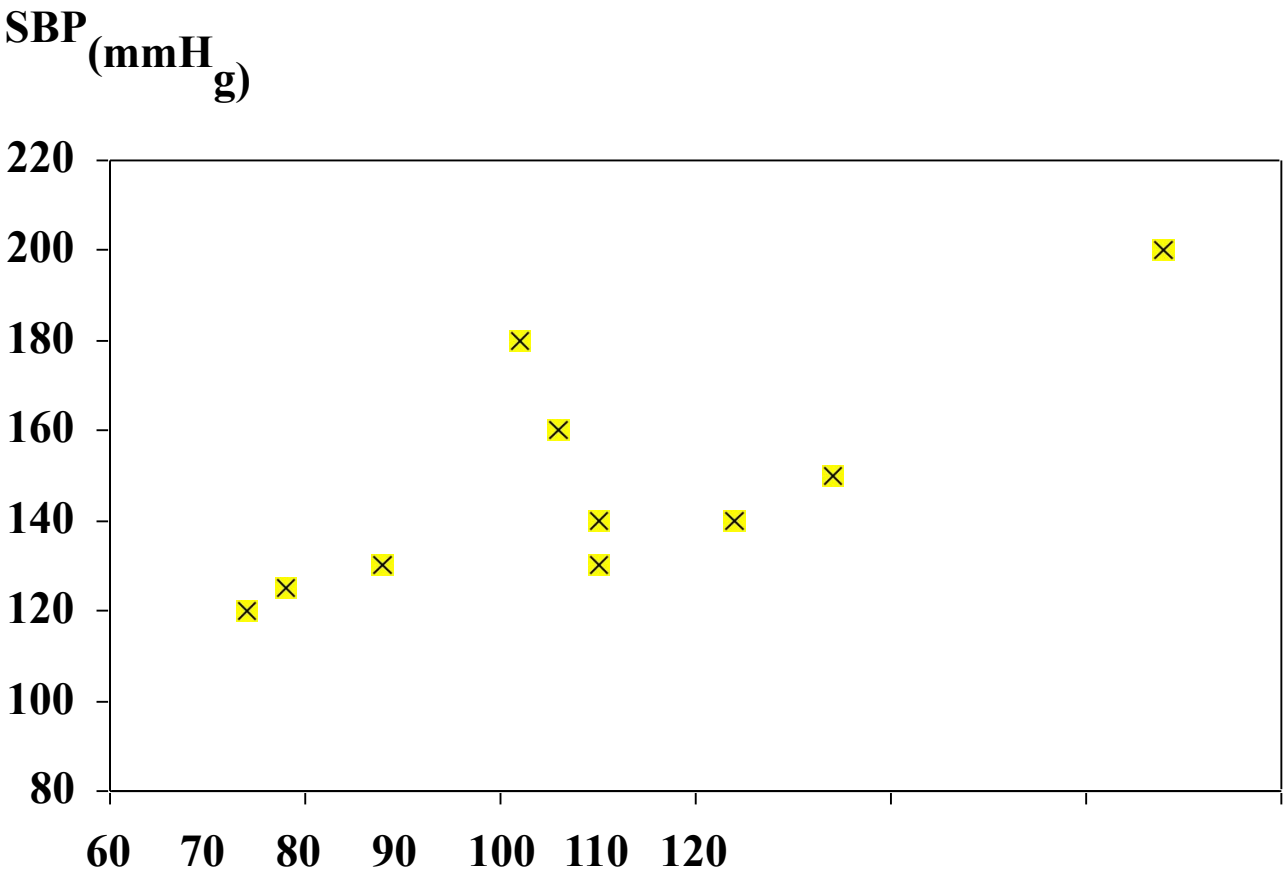


This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Example

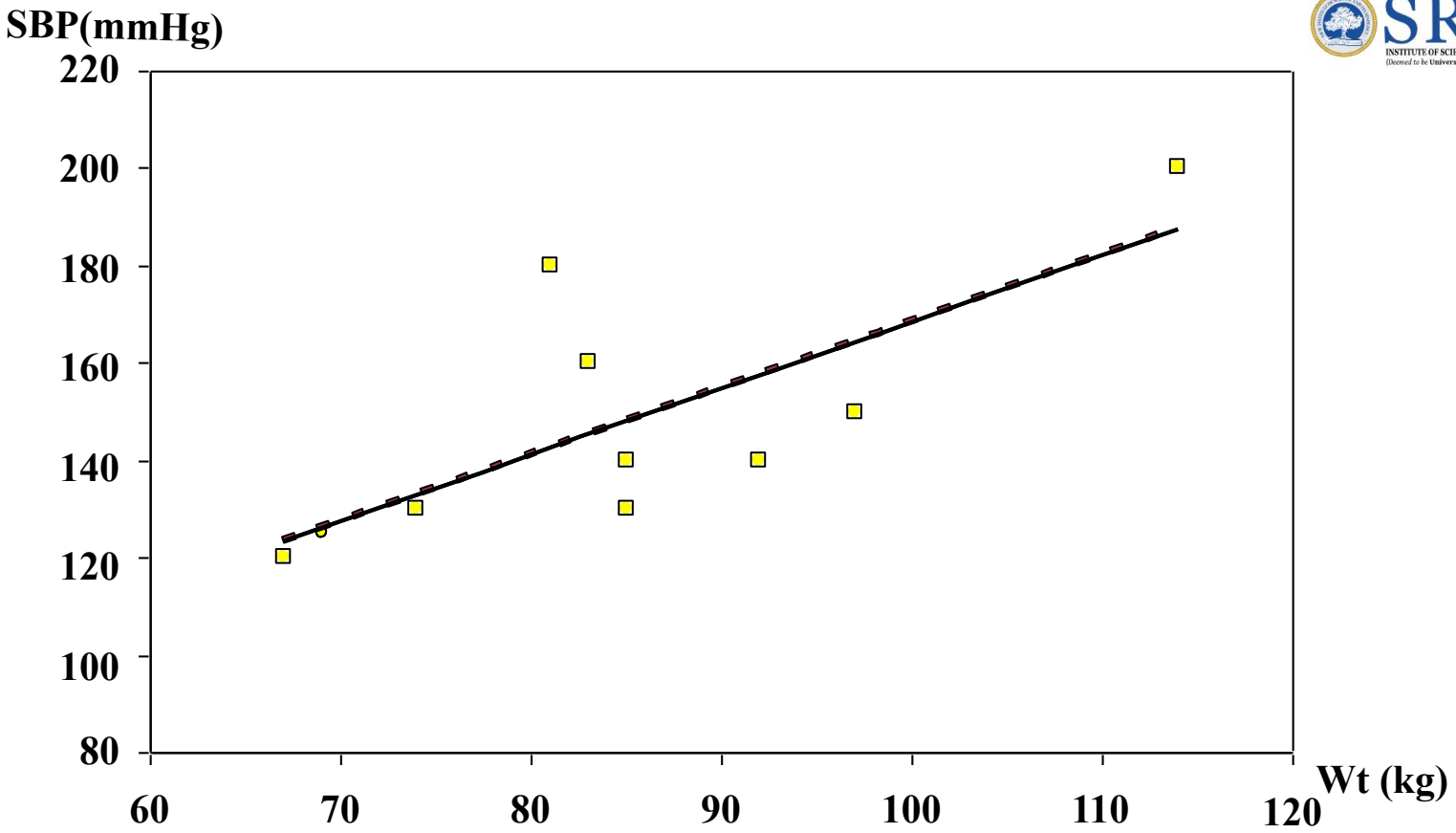
Wt. (kg)	67	69	85	83	74	81	97	92	114	85
SBP mHg)	120	125	140	160	130	180	150	140	200	130

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.



Scatter diagram of weight and systolic blood pressure

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.



Scatter diagram of weight and systolic blood pressure

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

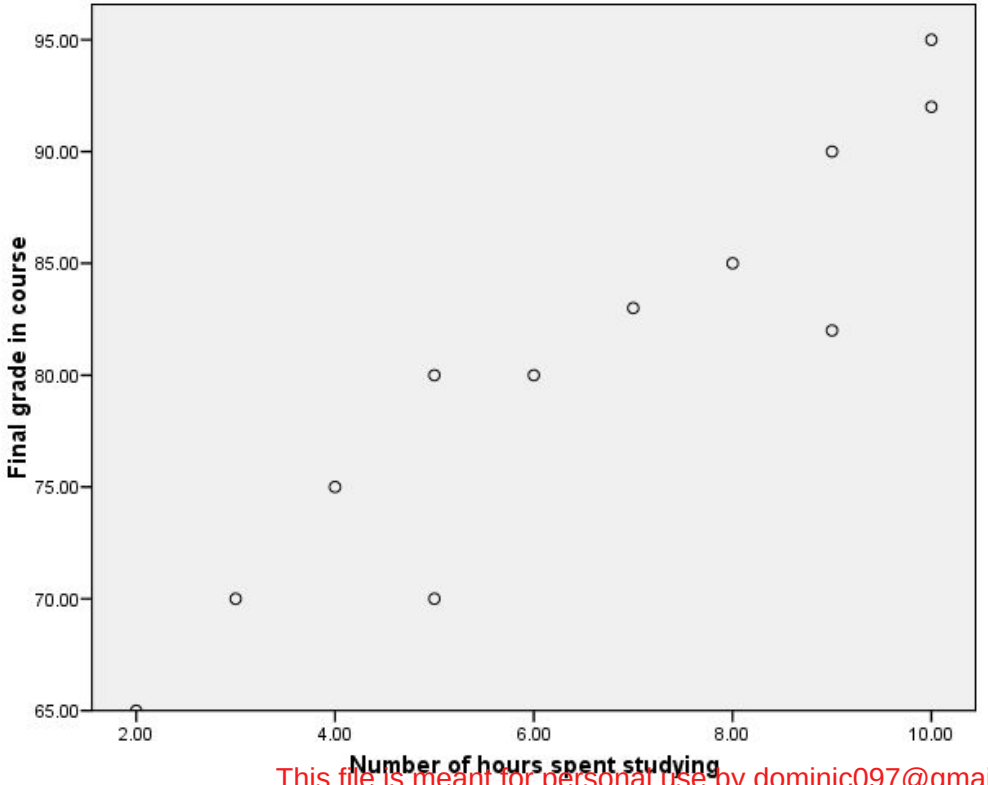
Scatter plots

The pattern of data is indicative of the type of relationship between your two variables:

- positive relationship
- negative relationship
- no relationship

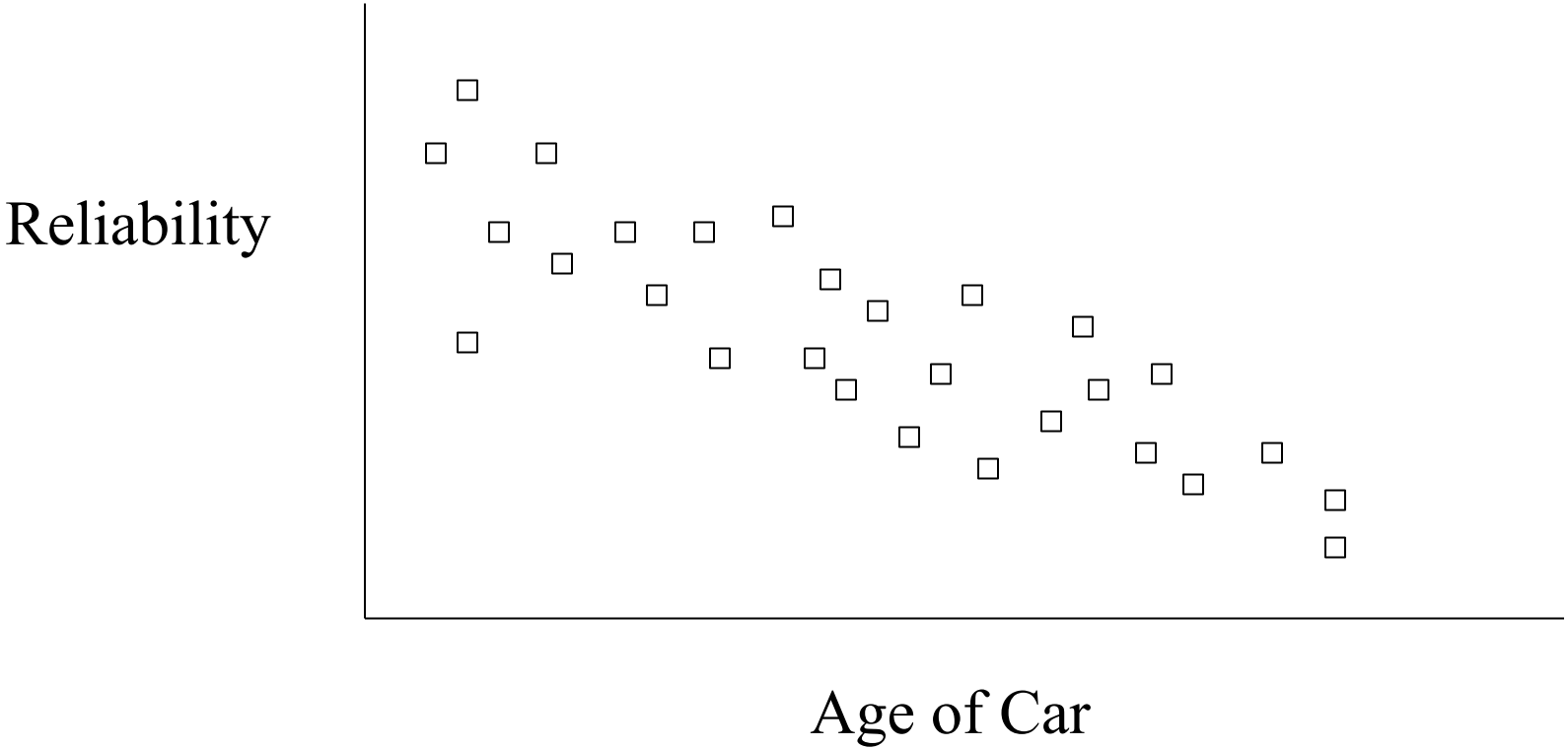
This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Positive relationship



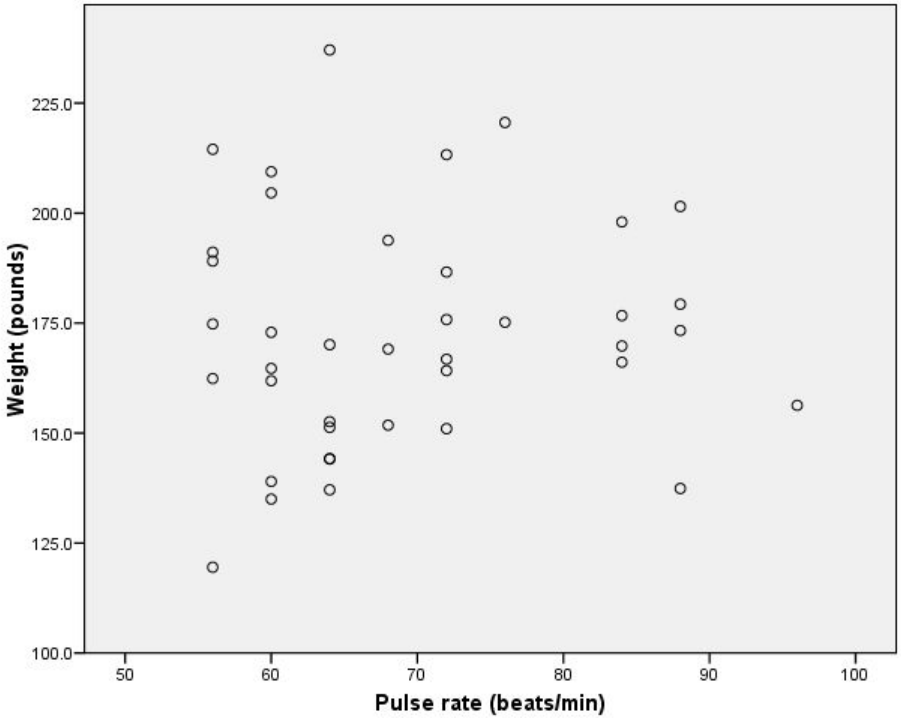
This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Negative relationship



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

No relation



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Correlation Coefficient

Statistic showing the degree of relation between two variables

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Simple Correlation coefficient (r)

- It is also called Karl Pearson's correlation or product moment correlation coefficient.
- Measures the nature and strength between two variables of the quantitative type.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

- Sign of r denotes the nature of association

- Value of r denotes the strength of association.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

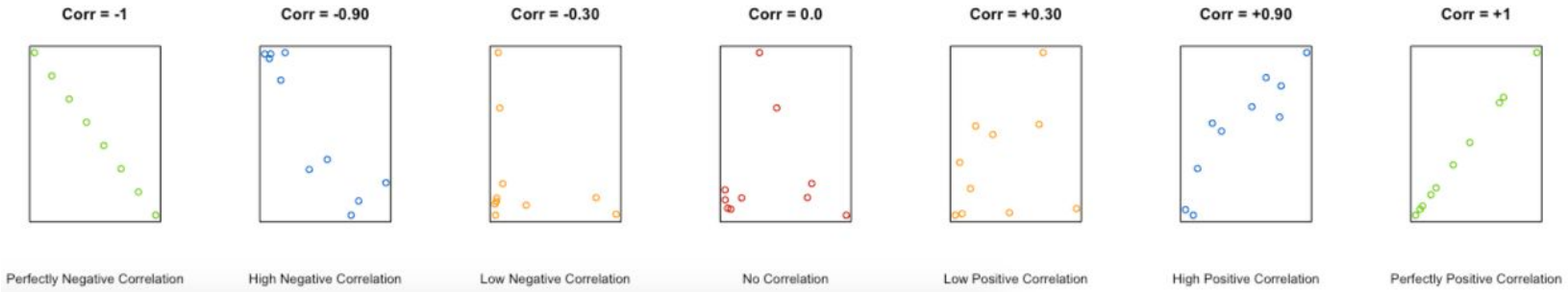
- If the sign is +ve this means the relation is direct (an increase in one variable is associated with an increase in the other variable and a decrease in one variable is associated with a decrease in the other variable).

- While if the sign is -ve this means an inverse or indirect relationship (which means an increase in one variable is associated with a decrease in the other).

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

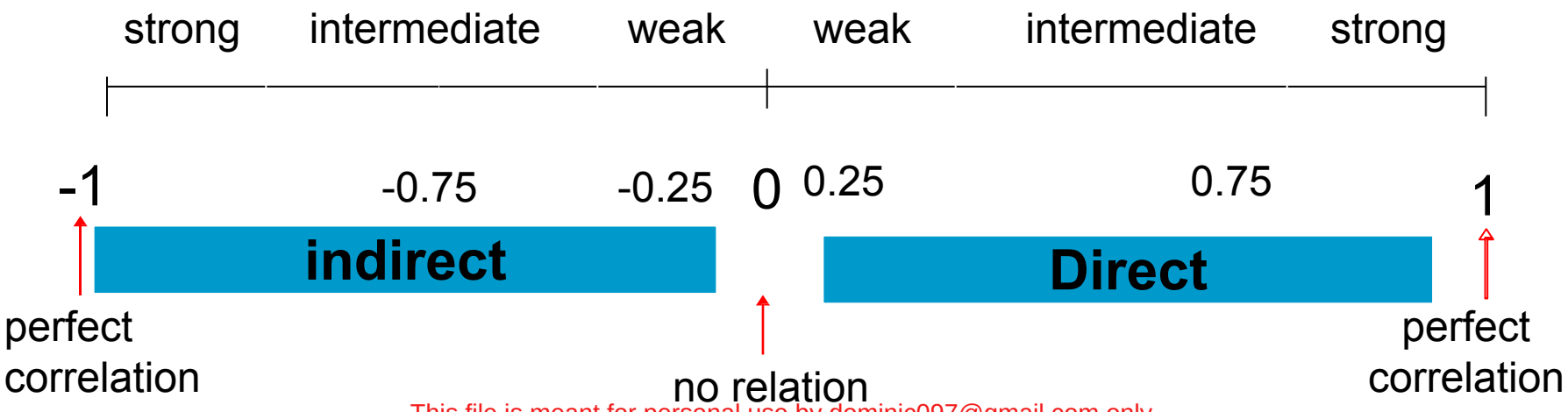
Correlation

- A positive correlation exists if variable increase/decrease simultaneously
- A negative correlation exists if one variable increases, while the other variable decreases



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

- The value of r ranges between (-1) and (+1)
- The value of r denotes the strength of the association as illustrated by the following diagram.



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

- If $r = 0$ this means no association or correlation between the two variables.
- If $0 < r < 0.25$ = weak correlation.
- If $0.25 \leq r < 0.75$ = intermediate correlation.
- If $0.75 \leq r < 1$ = strong correlation.
- If $r = 1$ = perfect correlation.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Correlation

- Correlation is normalized form of covariance

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sigma_x \sigma_y}$$

where σ_x is the standard deviation of X and σ_y is the standard deviation of Y

$$r = \frac{\sum(X-\overline{X})(Y-\overline{Y})}{\sqrt{\sum(X-\overline{X})^2}\sqrt{(Y-\overline{Y})^2}}$$

Where, $\overline{X}$ = mean of X variable

$\overline{Y}$ = mean of Y variable

- The two terms conceptually they are almost identical

This file is meant for personal use by sonibin097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Suppose we have marks of 5 students in **Math** and **Science**. Find the relation between the mark scoring ability of the students.

Student	Math (X)	Science (Y)
1	80	85
2	90	95
3	70	65
4	60	70
5	85	80

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Student	X	Y	$X-\bar{X}$	$Y-\bar{Y}$	$(X-\bar{X})(Y-\bar{Y})$
1	80	85	3	6	18
2	90	95	13	16	208
3	70	65	-7	-14	98
4	60	70	-17	-9	153
5	85	80	8	1	8
Σ					**485**

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\mathrm{Cov}(X,Y) = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{n}$$

$$\mathrm{Cov}(X,Y) = \frac{485}{5} = 97$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Student	$(X - \bar{X})^2$	$(Y - \bar{Y})^2$
1	9	36
2	169	256
3	49	196
4	289	81
5	64	1
Σ	**580**	**570**

$$\sigma_X = \sqrt{\frac{580}{5}} = \sqrt{116} \approx 10.77$$

$$\sigma_Y = \sqrt{\frac{570}{5}} = \sqrt{114} \approx 10.68$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$r = \frac{97}{10.77 \times 10.68}$$

$$r = \frac{97}{114.9} \approx 0.844$$

Correlation = 0.84 → Strong positive relationship between Math and Science marks.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Covariance and correlation

Covariance

- tells whether two variables are related by measuring how the variables change in relation to each other
- Gives the direction of the relationship

Correlation

- Similar to covariance
- Its tells strength and how strong is the relationship

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

DO IT

A café owner wants to know if higher temperatures increase ice cream sales. He records data over **6 days**:

Day	Temperature (°C)	Ice Cream Sales (units)
1	20	40
2	22	45
3	25	60
4	27	65
5	30	70
6	32	85

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Final Results

Sample Covariance: ≈ 75.0

Correlation (r): ≈ 0.99

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Permutation and Combination

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Permutation

A permutation is an arrangement of objects in a specific order. The sequence or order of the objects is important.

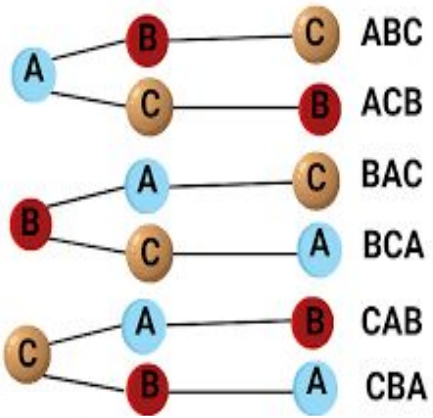
Formula:

$$^{n}P_{r} = \frac{n!}{(n-r)!}$$

n : Total Number of items.

r : Number of items need to select from n numbers

P(n,r): Number of permutations of n objects taken r at a time.



Key Points:

- **Order Matters:** Changing the order of objects creates a different permutation.
- **Examples:** Arranging books on a shelf, creating passwords.
- **Applications in Data Science:** Used in probability calculations, algorithm design, and scenario analysis.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Permutation

A website requires users to create a 4-character password using the digits 0-9. The digits cannot be repeated. How many different passwords can be generated?

- **Solution:**
  - Here, the order of digits matters because "1234" is a different password than "4321."
  - This is a permutation problem where n=10(total digits available) and r=4 (digits in the password).

Answer : Total possible passwords: 5040.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

DO IT

1) A project manager has 5 distinct tasks to assign to 5 time slots in a day. Each task must be completed in one of the slots, and the order of tasks is crucial. How many different ways can the tasks be scheduled?

Solution :

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

This means there are **120 different ways** to arrange the 5 tasks into the 5 available time slots.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

DO IT

2) You have 4 distinct features (F1, F2, F3, F4) that you want to use for training a machine learning model. You are interested in analyzing different ways to order these features and how their order might affect the model's performance. How many different permutations of these features are possible?

Solution:
This is a permutation problem where you are arranging all 4 features in different orders. The number of permutations is calculated as:

$$P(4,4)=4!=24$$

Total Permutations: 24 different ways to order the 4 features.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Combination

A combination refers to a selection of items where the order does not matter. Unlike permutations, changing the order of selected items does not create a new combination.

Formula:

The number of combinations of n items taken r at a time is given by:

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

n: Total number of items.
r: Number of items to choose.

Key Points:

- **Order Doesn't Matter:** "AB" is the same as "BA".
- **Examples:** Choosing a subset of students from a class, selecting a team members for a game.
- **Applications in Data Science:** Feature selection, experimental design, and data sampling.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Example:

Imagine you have 5 distinct books (A, B, C, D, E) and you want to select 3 of them. The number of possible combinations is calculated as:

Solution :

Total Combinations: 10 ways to choose 3 books from 5.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

1) You are organizing a team-building event and need to select 3 team members from a pool of 8 employees. The order in which the team members are selected does not matter.

Solution:

This is a combination problem where you are choosing 3 out of 8 employees, and the order does not matter. You can use the combination formula:

Total Combinations: 56 ways to select 3 team members from 8 employees.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

2) You are working on a machine learning project and have 15 distinct features (variables) available in your dataset. You want to select a subset of 5 features to use in your model. The order in which the features are selected does not matter.

Total Combinations: There are **3,003** different ways to select 5 features from the 15 available features for use in your machine learning model.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Probability

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Probability

- Probability is a measure for the likelihood of occurrence of an event

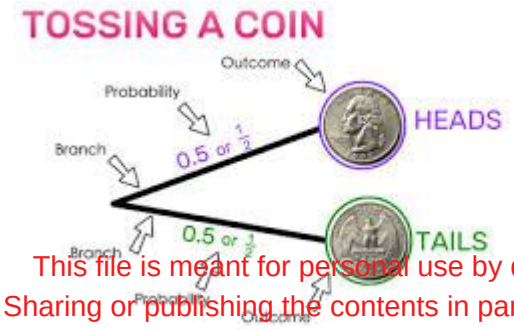
$$\text{Probability of an event} = \frac{\text{Number of favourable outcomes}}{\text{Number of all possible outcomes}}$$

- The probability of any event lies in between 0 to 1
- The sum of all possible events of an experiment is 1
- Probability of an event A is denoted by P(A)

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Terminologies in Probability

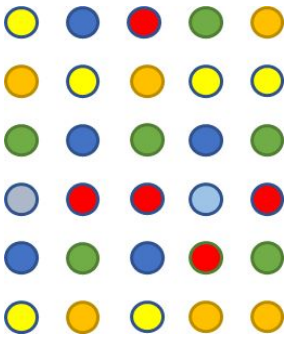
- Outcome : The result of a single trial of an experiment.
- Event : It is a set of one or more possible outcomes.
- Experiment : An experiment is a process or trial that produces one of several possible outcomes.
- Sample Space : set of all possible outcomes of an experiment or collection of event



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

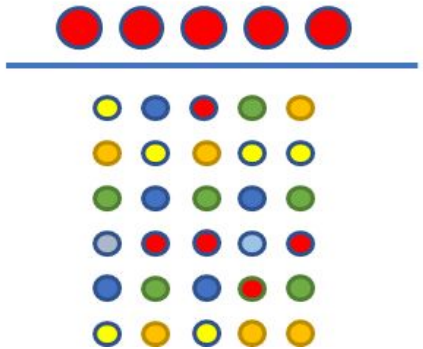
Probability

$$\text{Probability of an event } P(A) = \frac{\text{Number of outcomes in A}}{\text{Number of all possible outcomes}}$$



All possible outcomes

Probability (Ball is red) =



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

PLEASE!! NOTE

- Probability of complement of an event A is 1 - P(A)
- Probability of an impossible event is 0, i.e. $P(\Phi) = 0$
- For any event B, $0 \leq P(B) \leq 1$
- If event A is a subset of event B ($A \subset B$), then $P(A) \leq P(B)$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Probability

Question:

The new vaccine is to be tested on several patients. There are 5 diabetic patients with same type of diabetes, 9 non-diabetic patients with same heart condition and 11 non-diabetic patients with same liver condition. One patient is randomly chosen. What is the probability that the patient randomly picked is diabetic?

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Probability

Solution:

Let A: event that a patient is chosen

Total number of patients = 5 + 9 + 11 = 25

Number of diabetic patients = 5

Probability that a diabetic patients is chosen, i.e. $P(A) = \dfrac{\text{Number of diabetic patients}}{\text{Total number of patients}}$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Probability

Solution:

$$P(A) = \frac{\text{Number of diabetic patients}}{\text{Total number of patients}} = \frac{5}{25} = 0.2$$

What is the probability of choosing non-diabetic patients is , i.e P(A')

The probability of choosing non-diabetic patients is , i.e P(A')

P(A') = 1 - P(A) = 1- 0.2 = 0.8

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Independence of Events

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Independent Events

Two events A and B are **independent** if the occurrence of one event does not affect the probability of the other event occurring.

Mathematical Representation:

$$P(A \cap B) = P(A) \cdot P(B)$$

Key Points:

- Independent events do not influence each other.
- The combined probability is the product of their individual probabilities.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Independent Events

Example: Tossing Two Coins

- Event A: Getting Heads on the first coin.
- Event B: Getting Tails on the second coin.
  The outcome of the first coin (Heads or Tails) does not influence the outcome of the second coin.
  - Probability:
    - $P(A)=\frac{1}{2}$
    - $P(B)=\frac{1}{2}$
    - $P(A \cap B)=P(\text{Heads on first coin and Tails on second coin})$ $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Independent event

Question:

Consider rolling a fair die twice :

A: First trial : Occurrence of a number greater than 3

B: Second trail: Occurence of a number lesser than 3

Are the events A and B independent? Why.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Independent event

Solution:

Here the events are:

A: Occurrence of a number greater than 3

B: Occurrence of a number lesser than 3

i.e., $A = \{4, 5, 6\}$ and $B = \{1, 2\}$

$P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{3}$

$P(A \cap B) = P(A).P(B) = \frac{1}{6}$

Thus, events A and B are independent.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

DO IT

A company is running two independent marketing campaigns:

1. **Campaign A**: A digital ad campaign.
2. **Campaign B**: A social media promotion.

The company wants to assess the probability that a customer will make a purchase after being exposed to these campaigns.

- **Probability of making a purchase after seeing Campaign A**: 0.4 (40%).
- **Probability of making a purchase after seeing Campaign B**: 0.3 (30%).

What is the probability that a customer will make a purchase if they are exposed to both Campaign A and Campaign B?

Solution:

Since the two campaigns are independent, the probability of making a purchase after exposure to both campaigns is the product of the individual probabilities:

1. **Probability of making a purchase after seeing Campaign A**: $P(A)=0.4$
2. **Probability of making a purchase after seeing Campaign B**: $P(B)=0.3$
3. **Probability of making a purchase after seeing both Campaigns (A and B)**:
   $P(\text{Purchase} \mid A \text{ and } B)=P(A)\times P(B)=0.4\times 0.3=0.12$

Therefore, the probability of a customer making a purchase after being exposed to both campaigns is **0.12** or **12%**.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Mutually Exclusive Event

Two events are said to be **mutually exclusive** if they cannot happen at the same time. The occurrence of one event prevents the occurrence of the other.

Mathematical Representation:

$$P(A \cap B)=0$$
$$P(A \cup B) = P(A) + P(B)$$

Where $A \cap B$ represents both events happening together, which is impossible for mutually exclusive events.

Example: Rolling a Die

- **Event A:** Rolling a 1 (Outcome: {1}).
- **Event B:** Rolling a 2 (Outcome: {2}).
  Since you cannot roll both a 1 and a 2 in a single roll of the die, these events are **mutually exclusive**.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

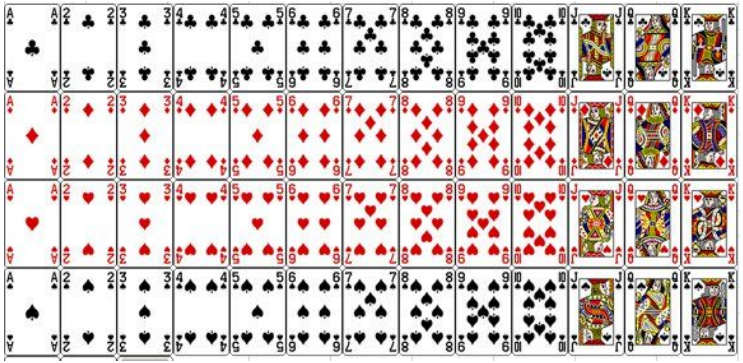
Mutually exclusive

Question:

A card is drawn from a pack of 52 cards. Let events A and B be:

A: Occurrence of a black card

B: Occurrence of a red card



Are the events A and B mutually exclusive? Why.

Solution:

Event A is a black card 26/52 and event B is the occurrence of a red card 26/52.

Implies, A ∩ B is empty. Thus the events A and B mutually exclusive.

This file is meant for personal use by dominic0976@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

DO IT

A retail company is analyzing the likelihood of customer purchases based on the time of day they shop. They identify two mutually exclusive segments for customers:

1. **Segment A**: Customers who shop in the morning (8 AM - 12 PM).
2. **Segment B**: Customers who shop in the afternoon (12 PM - 4 PM).

It is known from past data that:

- **Probability of a customer shopping in the morning**: 0.45 (45%).
- **Probability of a customer shopping in the afternoon**: 0.40 (40%).

These two segments are mutually exclusive because a customer can only be in one segment at a time; they cannot shop in both time periods on the same visit. What is the probability that a randomly selected customer will shop either in the morning or the afternoon?

Solution:

Since the two segments are mutually exclusive, the probability of a customer shopping in either time period is the sum of the individual probabilities:

1. **Probability of shopping in the morning**: P(A)=0.45P(A) = 0.45P(A)=0.45
2. **Probability of shopping in the afternoon**: P(B)=0.40P(B) = 0.40P(B)=0.40
3. **Probability of shopping either in the morning or in the afternoon**:
   P(Morning or Afternoon)=P(A)+P(B)=0.45+0.40=0.85

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

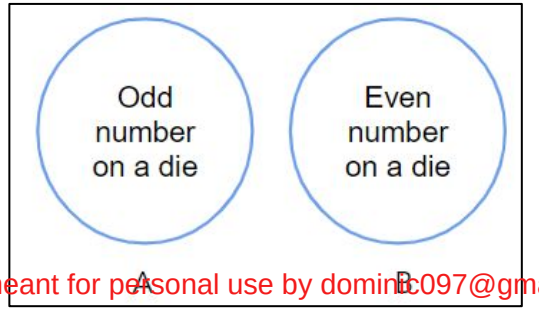
Therefore, the probability that a customer will shop either in the morning or in the afternoon is **0.85** or **85%**.

PLEASE!! NOTE

Independence and mutually exclusive events

- If A and B are independent events with P(A) and P(B) both non-zero, then A and B cannot be mutually exclusive

- If A and B are mutually exclusive with P(A) and P(B) both non-zero, then A and B cannot be independent



This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

PLEASE!! NOTE

Union ($A \cup B$): "OR"

- The probability that event A **or** event B (or both) will occur.
- *Example:* The probability of rolling a 2 **or** a 4 on a single die.

Intersection ($A \cap B$): "AND"

- The probability that event A **and** event B will **both** occur.
- *Example:* The probability of rolling an even number **and** a number greater than 4.

"At least"

- Means that number or more.
- *Example:* The probability of rolling **at least** a 5 means rolling a 5 or a 6.

"At most"

- Means that number or less.
- *Example:* The probability of rolling **at most** a 2 means rolling a 1 or a 2.

"Only"

- Refers to a single, specific outcome.
- *Example:* The probability of rolling **only** a 3.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

You roll a single six-sided die.

1) What is the probability of rolling an even number AND a number less than 4?
2) What is the probability of rolling a 5 OR a 6?

Solution:

- Sample space: $\{1, 2, 3, 4, 5, 6\}$
- **Probability of rolling an even number AND a number less than 4:**
  - The only number that satisfies both conditions is 2.
  - Favorable outcomes: 1
  - $$P(\text{even and } < 4) = \frac{1}{6}$$
- **Probability of rolling a 5 OR a 6:**
  - Favorable outcomes: 1 (for 5) + 1 (for 6) = 2
  - $$P(5 \text{ or } 6) = \frac{2}{6} = \frac{1}{3}$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

You roll two standard six-sided dice and add their numbers together. What is the probability of getting a sum of at least 10?

Solution:

Total number of possible outcomes: 6×6=36.

Probability of getting a sum of at least 10:

- The sums of 10, 11, or 12 are favorable.
- Sum of 10: (4,6), (5,5), (6,4) -> 3 outcomes
- Sum of 11: (5,6), (6,5) -> 2 outcomes
- Sum of 12: (6,6) -> 1 outcome
- Total favorable outcomes: 3+2+1=6.
- P(atleast 10) = 6/36 = 0.167

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

A jar contains 5 red, 4 blue, and 1 green marble. You randomly select three marbles from the jar . What is the probability of selecting atmost one red marble?

Solution:

Probability of no red marble = P(0 red) = 5/10 * 4/9 * 3 / 8 = 60/720 = **1/12**

Probability of one red marble:

1. Red, non red, non red = 5/10 * 5/9 * 4/8 = 100/720

2. Non red, red, non red = 5/10 * 5/9 * 4/8 = 100/720

3. Non red, non red, red = 5/10 * 4/9 * 5 / 8 = 100/720

Probability of exactly on red = (100+100+100)/720 = **5/12**

Probability of utmost one head = 1/12 + 5/12 = 6/12 = 1/2

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Marginal Probability

- Marginal probability refers to the probability of a single event occurring, regardless of the outcomes of other related events.
- It is obtained by summing (or integrating) the joint probabilities over all possible outcomes of the other variables.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Example:

- Suppose you have the probability of buying a product (event A) given different income levels (event B). The marginal probability P(A) is the total probability of buying a product, considering all income levels.

Product Bought	High Income	Low Income	Total (Marginal Probability)
Yes	0.3	0.2	0.5
No	0.1	0.4	0.5

Visualization:

- In a contingency table, marginal probabilities are the totals for each row or column.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

DO IT

A company tracks employee performance and job satisfaction levels. Each employee is categorized based on their job satisfaction level and their performance rating. The company has categorized employees into three performance levels (High, Medium, Low) and three satisfaction levels (Satisfied, Neutral, Dissatisfied). Here is the data from a survey of 300 employees:

- **High Performance:** 100 employees
  - **Satisfied:** 70
  - **Neutral:** 20
  - **Dissatisfied:** 10
- **Medium Performance:** 150 employees
  - **Satisfied:** 80
  - **Neutral:** 40
  - **Dissatisfied:** 30
- **Low Performance:** 50 employees
  - **Satisfied:** 10
  - **Neutral:** 15
  - **Dissatisfied:** 25

Questions:

1. **Find the marginal probability that a randomly selected employee has a High Performance rating.**
2. **Find the marginal probability that a randomly selected employee is Satisfied.**

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Solution:

1. Marginal Probability of High Performance:

To find the marginal probability of an employee having a High Performance rating, sum the number of employees with High Performance and divide by the total number of employees:

P(High Performance) = Number of High Performance Employees/ Total Number of Employees
= 0.333

2. Marginal Probability of Satisfaction:

To find the marginal probability of an employee being Satisfied, sum the number of employees who are Satisfied across all performance levels and divide by the total number of employees:

P(Satisfied) = Number of Satisfied Employees/ Total Number of Employees
= 0.533

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Joint Probability

- **Joint Probability** refers to the probability of two (or more) events occurring simultaneously.
- It is denoted as $P(A \cap B)$, where A and B are two events.
- Represents the likelihood of both events happening together.

Example:

- **Event A**: Drawing a red card from a deck $P(A)=26/52=0.5$
- **Event B**: Drawing a King $P(B)=4/52=0.077$
- **Joint Probability** (Drawing a red card and a King): $P(A \cap B)=0.5 \times 0.077=0.0385$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Conditional Probability

- The conditional probability of an event A is the probability that the event will occur given the knowledge that an event B has already occurred

- It is denoted by,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

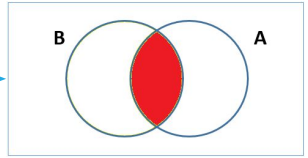
This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

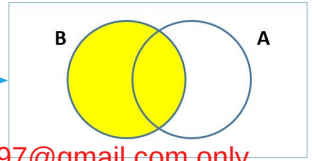
Conditional Probability

- The conditional probability of an event A is the probability that the event will occur given the knowledge that an event B has already occurred

- It is denoted by,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$





This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Conditional probability

Question:

A pair of fair dice is rolled. If the product of numbers that appear is 6, find the probability that the second die shows an even number?

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Conditional probability

Solution:

Second Dice \ First Dice	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Let A: the event of getting the product as 6

The ways A can occur: {(1,6), (2,3), (3,2), (6,1)}

Let B: the event that the second die shows an even number

The ways B can occur: {(1,2), (1,4), (1,6), (2,2), (2,4), (2,6), (3,2), (3,4), (3,6), (4,2), (4,4), (4,6), (5,2), (5,4), (5,6), (6,2), (6,4), (6,6)}

Thus, the event that product of the die is 6 and number 2 appears on the dice is $A \cap B$.

$$A \cap B = \{(1,6), (3,2)\}$$

The total number of samples is 36.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Conditional probability

Solution:

The required probability is

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)}$$

$$P\big(\text{number on the second die is even} \mid \text{the product is } 6\big) = \frac{\text{P (product is 6 and number on the second die is even)}}{\text{P(the product is 6)}}$$

$$P\big(\text{number on the second die is even} \mid \text{the product is } 6\big) = \frac{\frac{2}{36}}{\frac{4}{36}} = \frac{2}{4} = 0.5$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

PLEASE!! NOTE

Let A and B be two events:

Addition theorem: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Multiplication theorem:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow P(A \cap B) = P(A \mid B).\, P(B)$$

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

$$\Rightarrow P(A \cap B) = P(B \mid A).\, P(A)$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Rules for computing Probability

- If events are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B)$$

- If there is commonality between events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- If the events are Independent

$$P(A \cap B) = P(A) \,.\, P(B)$$

- If A and B are not independent, you must use conditional probability:

$$P(A \cap B) = P(A) \times P(B \mid A)$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Bayes' Theorem

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Bayes' theorem

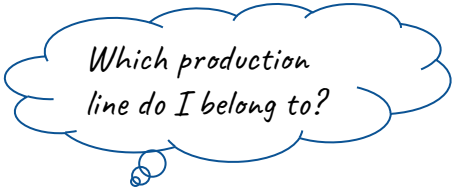
- Conditional probability is the likelihood of an event given that another event has occurred
- Bayes theorem provides a way to update the probability based on the new information
- It is completely based on the conditional probability
- Also known as Bayes' Rule or Bayes' law

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Bayes' theorem

Suppose L_1, L_2 and L_3 are the production lines of an industry and D is the event that the article manufactured is defective. Given the article is defective, we need to find the probability that it was produced on which line.

$$P(L_i \mid D) = \frac{P(L_i).P(D|L_i)}{\sum_{j=1}^{3} P(L_j).P(D|L_j)} \quad \text{for all } i = 1, 2, 3$$

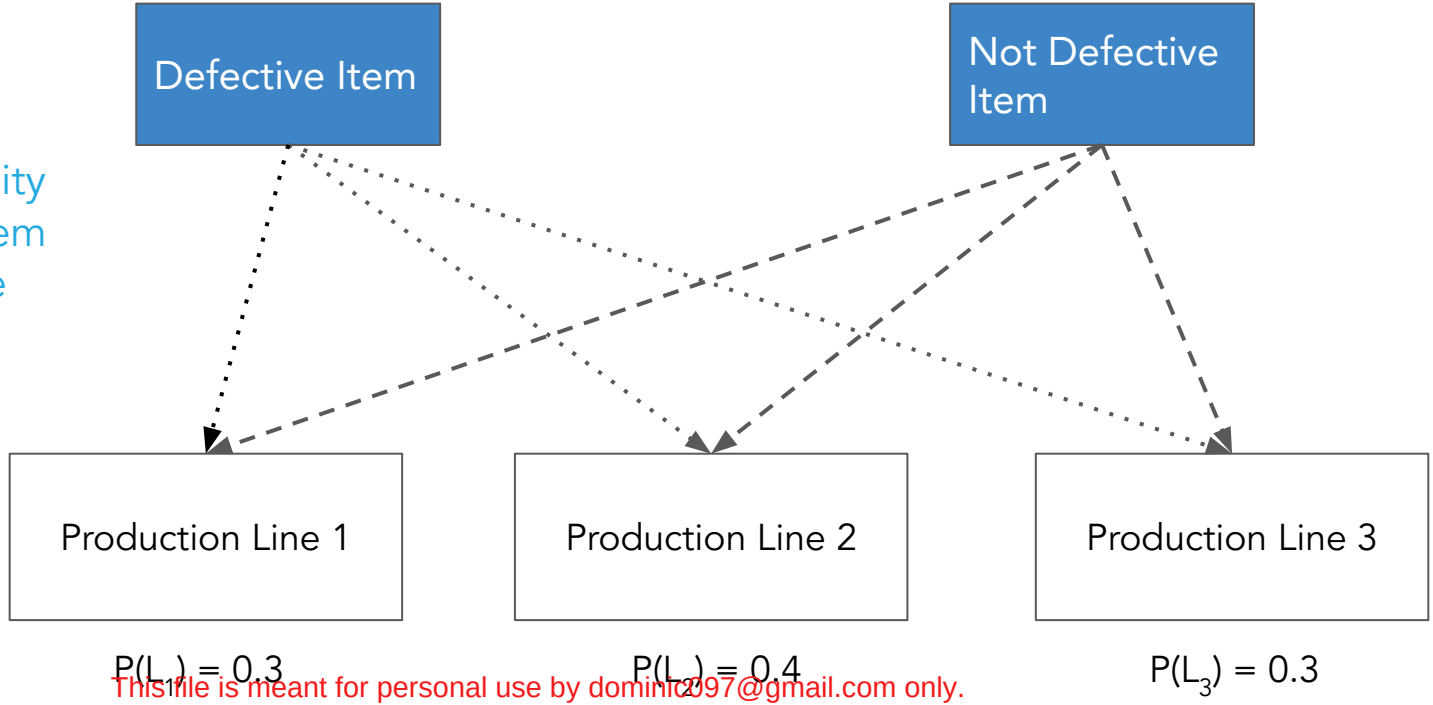


Defective Item

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Bayes' theorem

What is the probability that the defective item chosen was from the production line 1?



This file is meant for personal use by dominic297@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Bayes' theorem

Suppose events $A_1, A_2, \ldots, A_n$ form a partition on a sample space Ω of a random experiment and B is any other event such that P(B) > 0 defined on Ω, then

Prior Probabilities

Likelihood

$$P(A_i \mid B) = \frac{P(A_i).P(B|A_i)}{\sum_{j=1}^{n} P(A_j).P(B|A_j)}$$

for all i = 1, 2,...,n

Posterior Probabilities

Evidence

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Conditions in Bayes Theorem

1) P(A|B) not equal to P(B|A)
2) There should be dependency between the event.
3) Marginal Probability can’t be zero

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Prior probabilities

- Consider three production lines L_1, L_2, L_3 manufacturing the same finished articles
- From the past experience, it is seen that the chance of selecting a production line is 0.3, 0.4, 0.3 respectively
- These probabilities are called the prior probabilities

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Posterior probabilities

- Consider three production lines L_1, L_2, L_3 manufacturing the same finished articles

- An article is chosen at random and is found to be defective

- The knowledge of the probability that the defective article is produced by L_1, L_2 or L_3 will help in improving the quality of the product line, i.e., $P(L_1 \mid \text{defective article})$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Likelihood

- Consider three production lines L_1, L_2, L_3 manufacturing the same finished articles

- Likelihood is the conditional probability that the article produced is defective given it is produced on one of the lines L_1, L_2, L_3. i.e. P(defective article | L_1)

- It specifies the chance of producing a defective article of the given production line

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Evidence

- Event D is that the article selected is defective, which is not used in the calculation of the prior probability
- In our example, the evidence is the probability of selecting a defective article produced on any of the production lines L_1, L_2 or L_3
- It is also known as marginal probability

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Bayes' theorem

Question:

In an armament production station, the explosion can occur due to short circuit, fault in the machinery, negligence of workers. From experience, the chances of these causes are 0.1, 0.3, 0.6 respectively. The chief engineer feels that an explosion can occur with probability:

- 0.3 if there is a short circuit
- 0.2 if there is a fault in the machinery
- 0.25 if the workers are negligent

Given that an explosion has occurred, determine the most likely cause of it?

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Bayes' theorem

Solution:

Let the events be defined as

A_1: the explosion can occur due to short circuit

A_2: the explosion can occur due to fault in the machinery

A_3: the explosion can occur due to negligence of workers

The prior probabilities are

$P(A_1) = 0.1$, $P(A_2) = 0.3$, $P(A_3) = 0.6$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Bayes' theorem

Solution:

Let B be the event there is an explosion.

Thus the probabilities are

P(there is an explosion| there is a short circuit) = $P(B \mid A_1) = 0.3$

P(there is an explosion| there is a fault in the machinery) = $P(B \mid A_2) = 0.2$

P(there is an explosion| the workers are negligent) = $P(B \mid A_3) = 0.25$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Bayes' theorem

Solution:

To find: the most likely cause of it given that an explosion has occurred

Thus, using the bayes theorem, to compute the probability that there is a short circuit given an explosion has occurred is computed as

$$P(A_1|B) = \frac{P(A_1).P(B|A_1)}{P(A_1).P(B|A_1) + P(A_2).P(B|A_2) + P(A_3).P(B|A_3)}$$

$$P(A_1|B) = \frac{0.1\times0.3}{0.1\times0.3+0.3\times0.2+0.6\times0.25} = \frac{1}{8} = 0.125$$

This file is meant for personal use by dominic297@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Bayes' theorem

Solution:

Similarly,

$$P(A_2|B) = \frac{0.3\times0.2}{0.1\times0.3+0.3\times0.2+0.6\times0.25} = \frac{1}{4} = 0.25$$

$$P(A_3|B) = \frac{0.6\times0.25}{0.1\times0.3+0.3\times0.2+0.6\times0.25} = \frac{5}{8} = 0.625$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Bayes' theorem

Solution:

Thus, we have

P(there is a short circuit | there is an explosion) = $P(A_1 \mid \mathbf{B}) = 0.125$

P(there is a fault in the machinery| there is an explosion) = $P(A_2 \mid \mathbf{B}) = 0.25$

P(the workers are negligent| there is an explosion) = $P(A_3 \mid \mathbf{B}) = 0.625$

Comparing the probabilities, $P(A_3 \mid \mathbf{B})$ has highest probability. It implies that the negligence of workers is the most likely cause of an explosion in the factory.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Summary

Terminologies	Meaning
Prior Probability	Probability of the event A (P(A))
Posterior Probability	Conditional probability of the event A given event B has occurred (P(A|B))
Likelihood	Conditional probability of the event B given event A has occurred (P(B|A))
Evidence	Probability of the event B (P(B))

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Naive Bayes' Theorem

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

What is Naive Bayes?

- Naive Bayes is a supervised machine learning algorithm used for classification tasks.
- It's based on **Bayes' theorem**, with a "naive" assumption of conditional independence.
- It's particularly popular for **text classification** and **spam filtering**.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

The "Naive" Assumption

- The "naive" part comes from the assumption that all features in a dataset are **independent** of each other.
- This means the presence of one feature does not affect the presence of any other feature.
- **Example:** If we are classifying a fruit as an "apple," Naive Bayes assumes that the features "red," "round," and "has a stem" are all independent of each other. In reality, this is often not true, but the algorithm still performs surprisingly well.

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Naive bayes

- The algorithm calculates this probability for each possible class and then predicts the class with the **highest posterior probability**.
- The denominator, P(Features), is constant for all classes, so we can ignore it for comparison.

Formula for Classification:

$$\text{Classifier} = \text{argmax}_{C_i}[P(C_i) \cdot \prod_{j=1}^{n} P(x_j|C_i)]$$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Naïve Bayes Classifier: Training Dataset

Class:

C1:buys_computer = ‘yes’

C2:buys_computer = ‘no’

Data to be classified:

X = (age <=30,

Income = medium,

Student = yes

Credit_rating =

Fair)

age	income	student	credit_rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

This file is meant for personal use by dominic0017@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Naïve Bayes Classifier: An Example

age	income	student	redit_rating	_com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

- $P(C_i)$: P(buys_computer = "yes") = 9/14 = 0.643
  P(buys_computer = "no") = 5/14= 0.357
- Compute $P(X|C_i)$ for each class
  P(age = "<=30" | buys_computer = "yes") = 2/9 = 0.222
  P(age = "<= 30" | buys_computer = "no") = 3/5 = 0.6
  P(income = "medium" | buys_computer = "yes") = 4/9 = 0.444
  P(income = "medium" | buys_computer = "no") = 2/5 = 0.4
  P(student = "yes" | buys_computer = "yes) = 6/9 = 0.667
  P(student = "yes" | buys_computer = "no") = 1/5 = 0.2
  P(credit_rating = "fair" | buys_computer = "yes") = 6/9 = 0.667
  P(credit_rating = "fair" | buys_computer = "no") = 2/5 = 0.4

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Problem continued…

- **X = (age <= 30 , income = medium, student = yes, credit_rating = fair)**

$P(X|C_i)$: P(X|buys_computer = “yes”) = 0.222 x 0.444 x 0.667 x 0.667 = 0.044

P(X|buys_computer = “no”) = 0.6 x 0.4 x 0.2 x 0.4 = 0.019

$P(X|C_i)*P(C_i)$: P(X|buys_computer = “yes”) * P(buys_computer = “yes”) = 0.028

P(X|buys_computer = “no”) * P(buys_computer = “no”) = 0.007

Therefore, X belongs to class (“buys_computer = yes”)

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Short goal: given weather observations, predict **Play? = Yes / No** using **Naive Bayes**

#	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Strong	Yes

Test instance: **Outlook = Sunny, Temp = Cool, Humidity = High, Wind = Strong**

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

For **Yes**:

- $P(\text{Outlook=Sunny} \mid \text{Yes}) = 1/6 \approx 0.1667$
- $P(\text{Temp=Cool} \mid \text{Yes}) = 3/6 = 0.5$
- $P(\text{Humidity=High} \mid \text{Yes}) = 2/6 \approx 0.3333$
- $P(\text{Wind=Strong} \mid \text{Yes}) = 2/6 \approx 0.3333$

For **No**:

- $P(\text{Outlook=Sunny} \mid \text{No}) = 3/4 = 0.75$
- $P(\text{Temp=Cool} \mid \text{No}) = 1/4 = 0.25$
- $P(\text{Humidity=High} \mid \text{No}) = 3/4 = 0.75$
- $P(\text{Wind=Strong} \mid \text{No}) = 2/4 = 0.5$

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

For Yes:

- Prior: $P(\text{Yes}) = 0.6$
- Likelihood: $(1/6) \times (3/6) \times (2/6) \times (2/6)$
  = $0.1666667 \times 0.5 \times 0.3333333 \times 0.3333333$
  = 0.009259259259259259 (product of conditionals)
- Multiply by prior: $0.6 \times 0.009259259259259259 = 0.005555555555555555$

For No:

- Prior: $P(\text{No}) = 0.4$
- Likelihood: $0.75 \times 0.25 \times 0.75 \times 0.5 = 0.0703125$
- Multiply by prior: $0.4 \times 0.0703125 = 0.028125$

Final prediction:

Given weather condition it is predicted "NO" to play outside as P(NO|x)>P(YES|x)

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Thank you!

Happy Learning :)

This file is meant for personal use by dominic097@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.